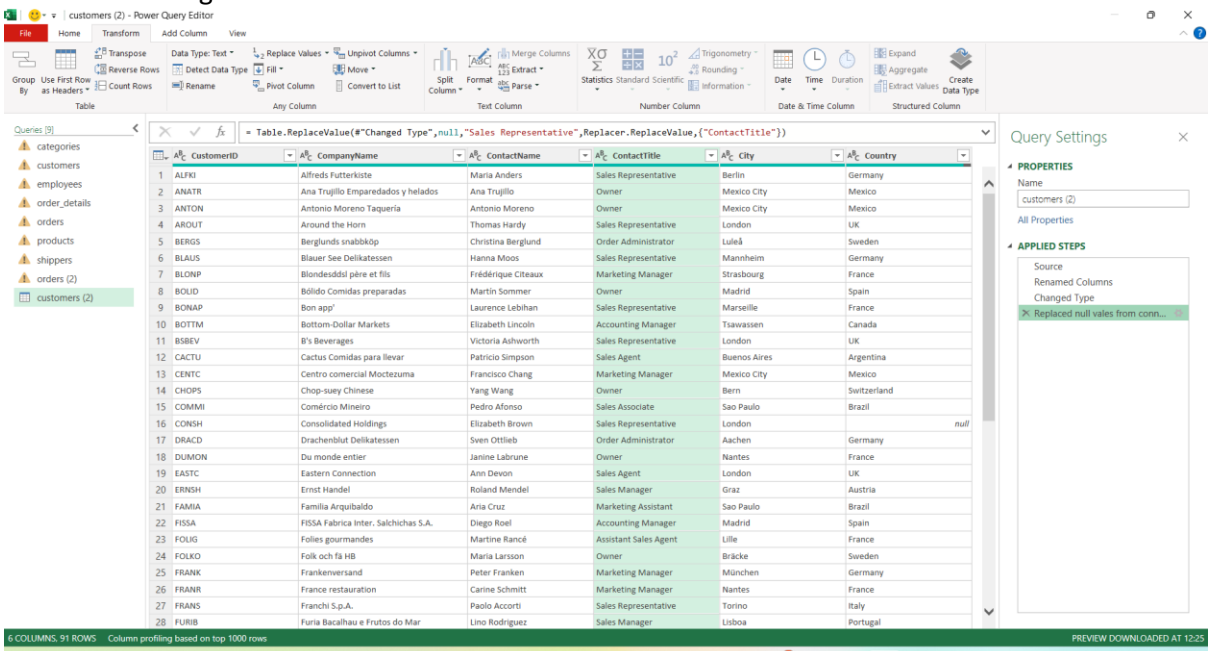
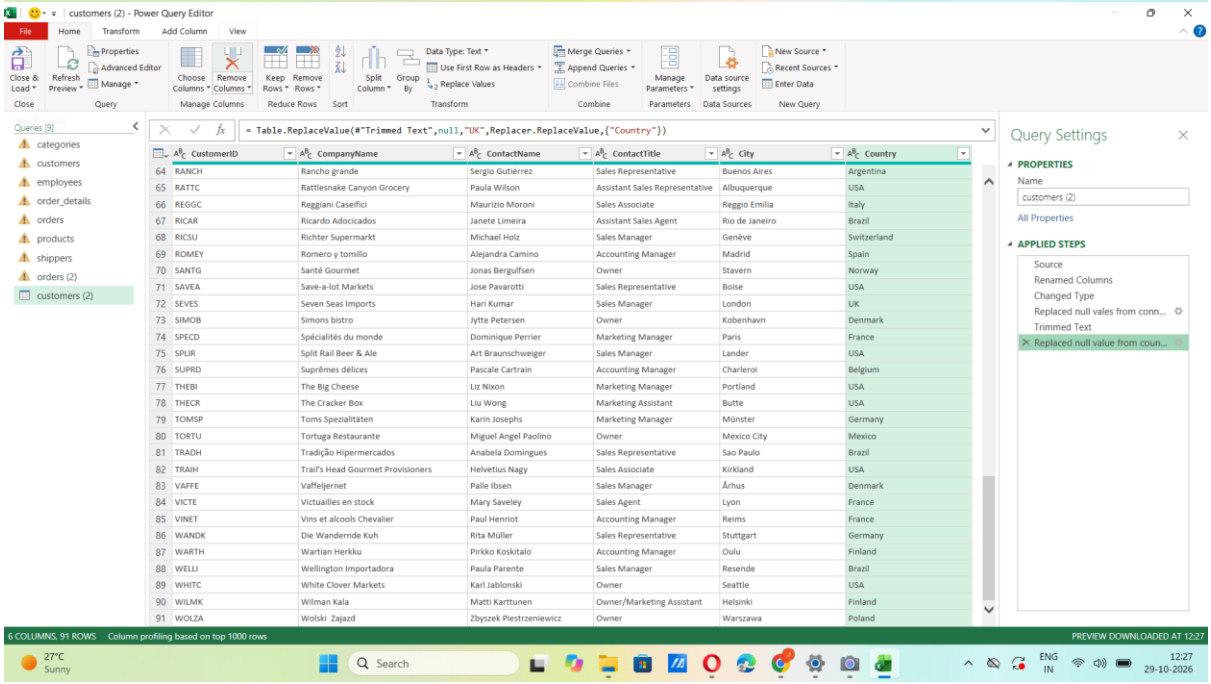
Mini Project – Data Cleaning and Transformation Observations

Step 1: Data Cleaning Using Power Query

1. Customer Table

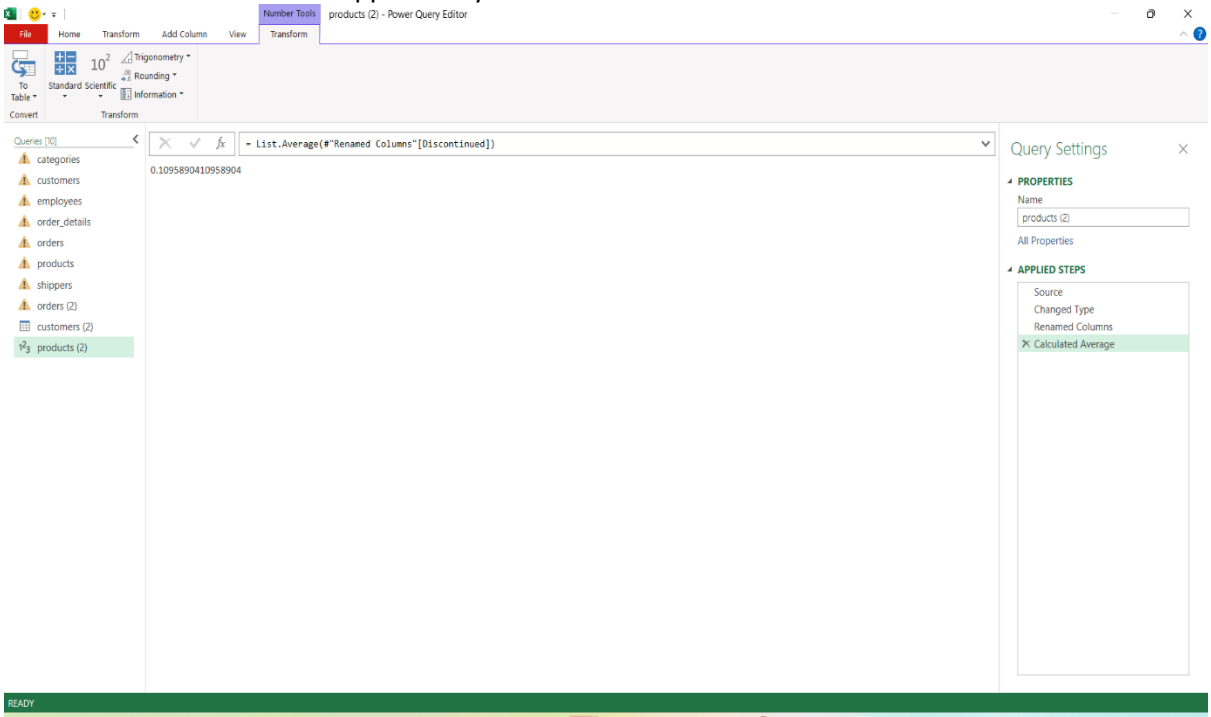
- Removed unnecessary spaces from text columns using the Trim function.
- Standardized text values into a consistent format.
- Changed column data types where required.
- Replaced null values using appropriate category-based values and the most frequently occurring values.

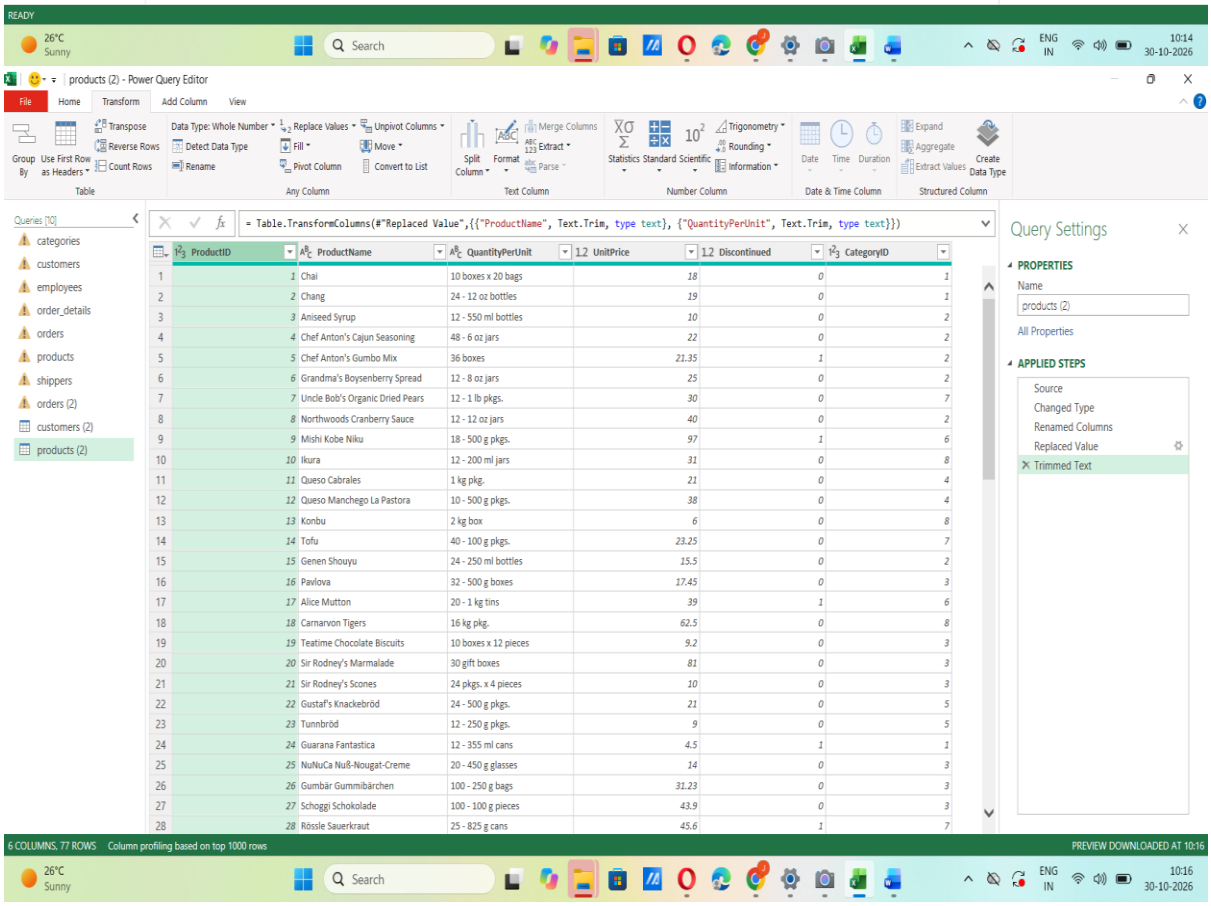




2. Product Table

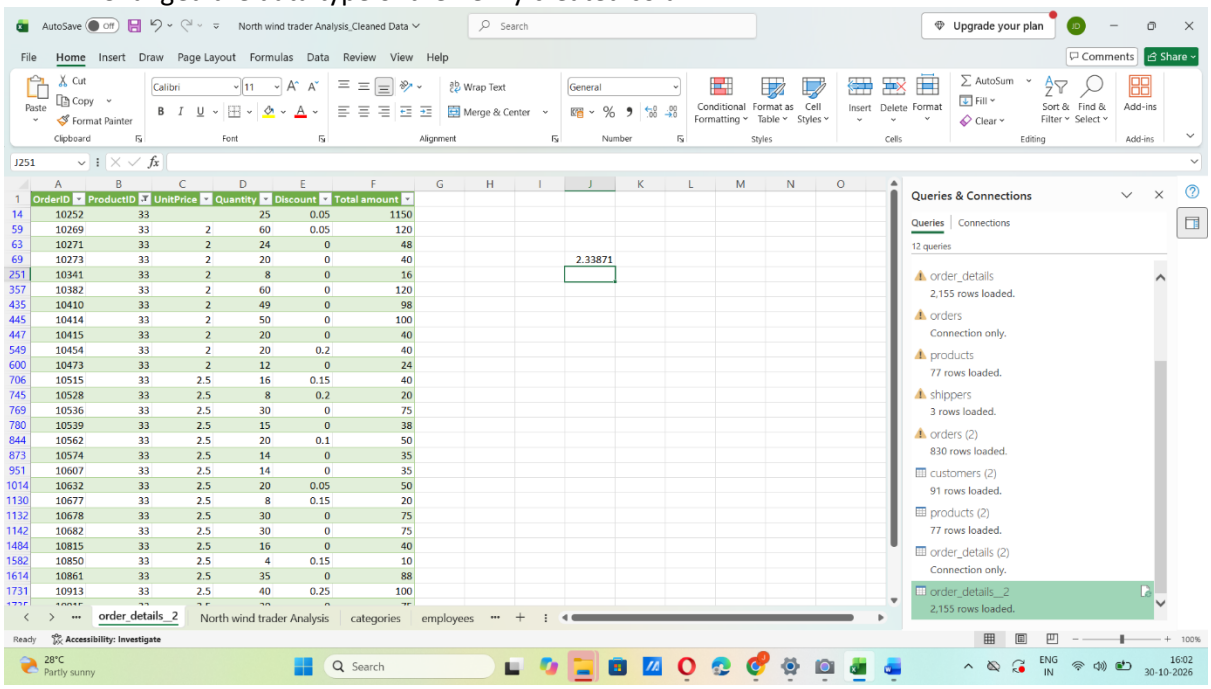
- Removed unnecessary spaces using the Trim function.
- Standardized text values and corrected data types.
- Calculated the average of the available values to handle missing values.
- Since the calculated average was approximately 0.109, replaced the null values with 0. This decision should be supported by the business context.



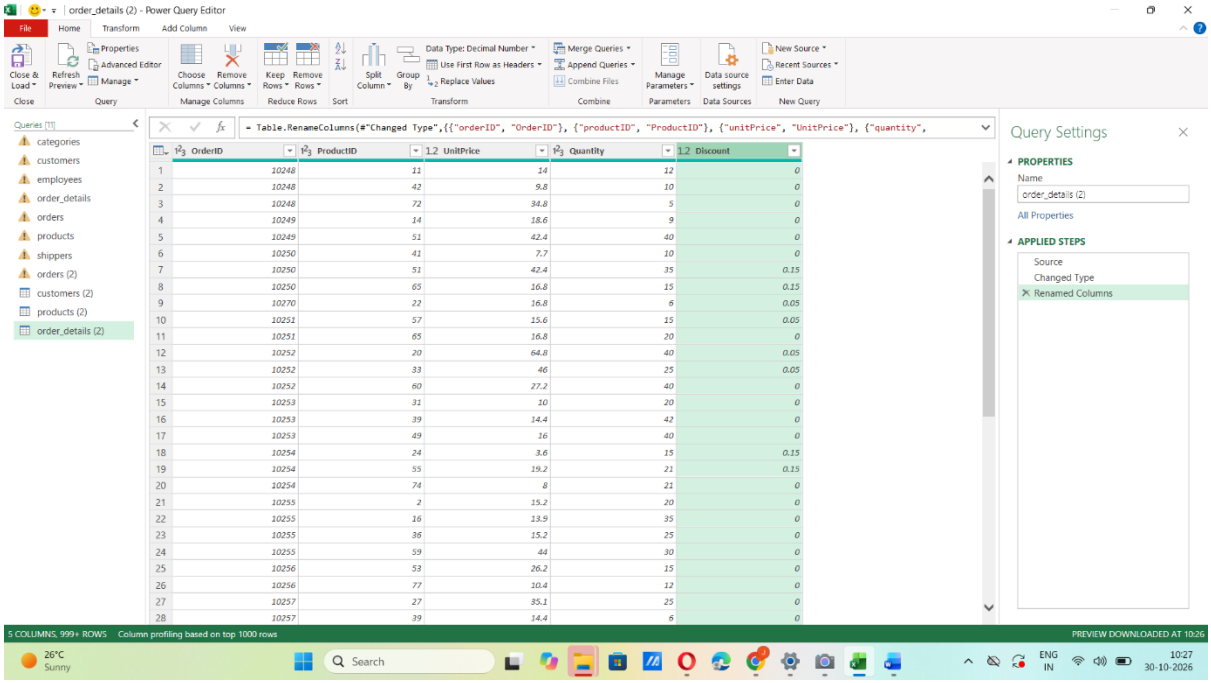


3. Northwind Trader Analysis Table

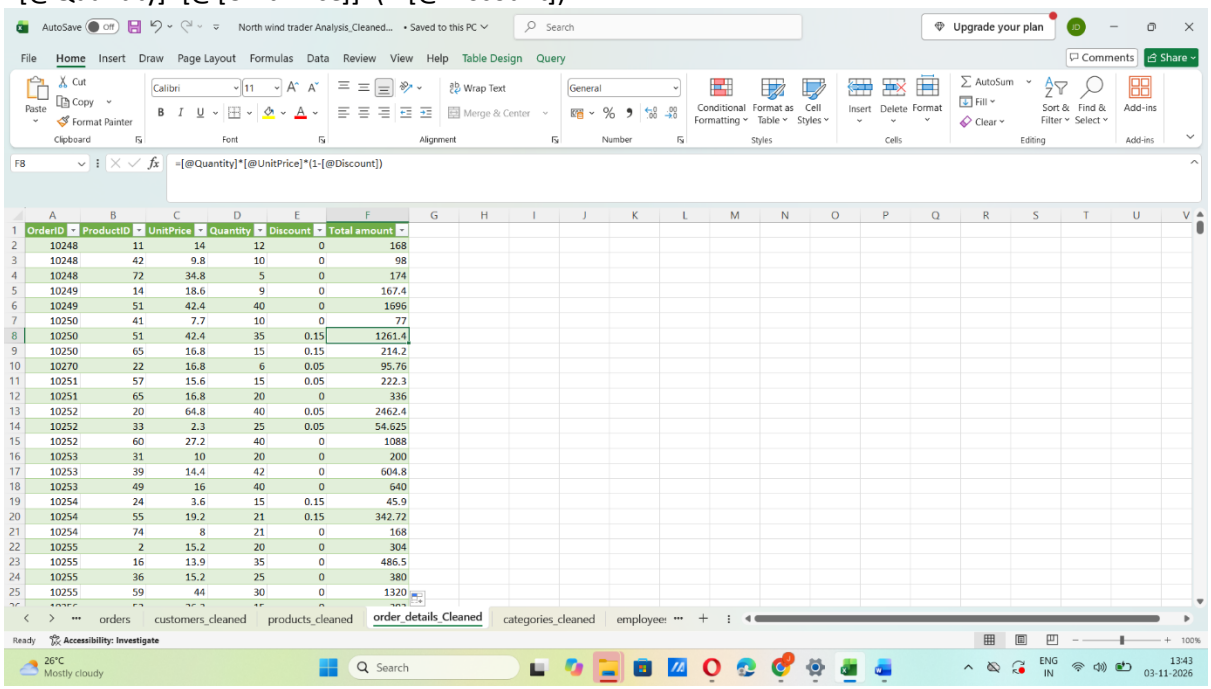
- Removed unnecessary spaces from the columns.
- Renamed column headers for better understanding.
- Replaced missing Unit Price values using the average Unit Price for the corresponding product category.
- Changed the data type of the newly created column.



Replaced missing Discount values using the average Discount for the corresponding product.



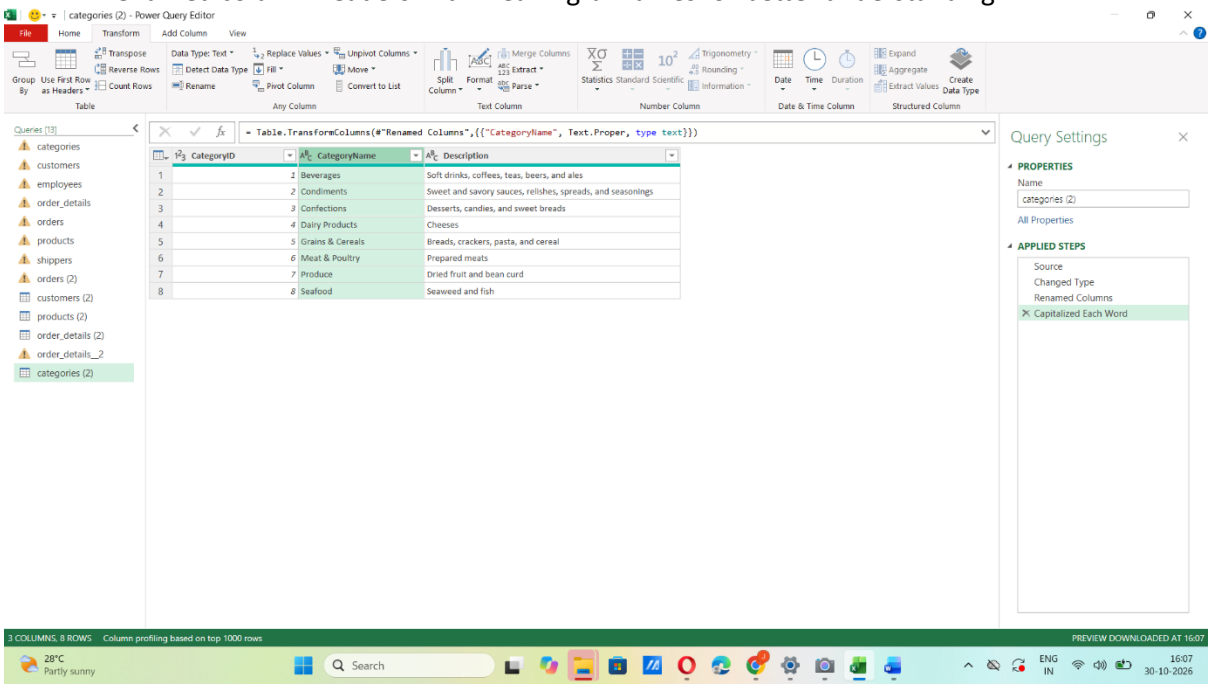
Created a custom column named Total Amount by calculating Quantity × Unit Price × (1 − Discount)..
=[@Quantity]*[@[Unit Price]]*(1-[@Discount])



OrderID	ProductID	UnitPrice	Quantity	Discount	Total amount
10248	11	14	12	0	168
10248	42	9.8	10	0	98
10248	72	34.8	5	0	174
10249	14	18.6	9	0	167.4
10249	51	42.4	40	0	1696
10250	41	7.7	10	0	77
10250	51	42.4	35	0.15	1261.4
10250	65	16.8	15	0.15	214.2
10270	22	16.8	6	0.05	95.76
10251	57	15.6	15	0.05	222.3
10251	65	16.8	20	0	336
10252	20	64.8	40	0.05	2462.4
10252	33	2.3	25	0.05	54.625
10252	60	27.2	40	0	1088
10253	31	10	20	0	200
10253	39	14.4	42	0	604.8
10253	49	16	40	0	640
10254	24	3.6	15	0.15	45.9
10254	55	19.2	21	0.15	342.72
10254	74	8	21	0	168
10255	2	15.2	20	0	304
10255	16	13.9	35	0	486.5
10255	36	15.2	25	0	380
10255	59	44	30	0	1320

4. Category Table

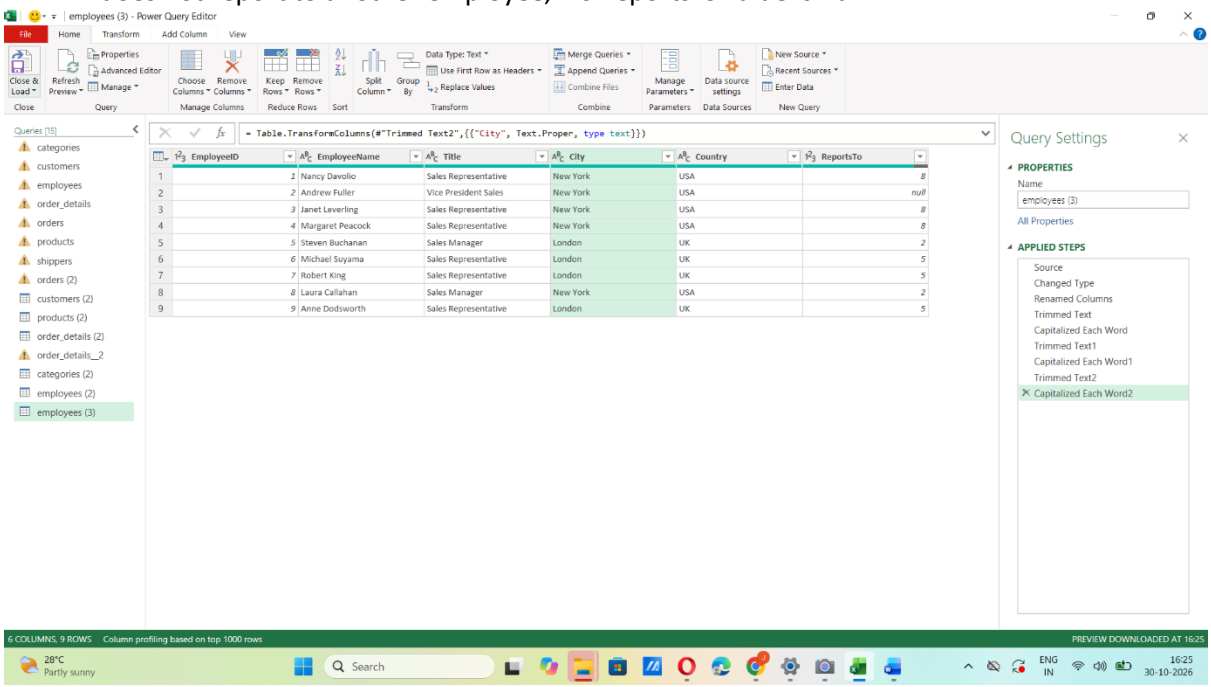
- Removed unnecessary spaces using the Trim function.
- Renamed column headers with meaningful names for better understanding.



CategoryID	CategoryName	Description
1	Beverages	Soft drinks, coffees, teas, beers, and ales
2	Condiments	Sweet and savory sauces, relishes, spreads, and seasonings
3	Confections	Desserts, candies, and sweet breads
4	Dairy Products	Cheeses
5	Grains & Cereals	Breads, crackers, pasta, and cereal
6	Meat & Poultry	Prepared meats
7	Produce	Dried fruit and bean curd
8	Seafood	Seaweed and fish

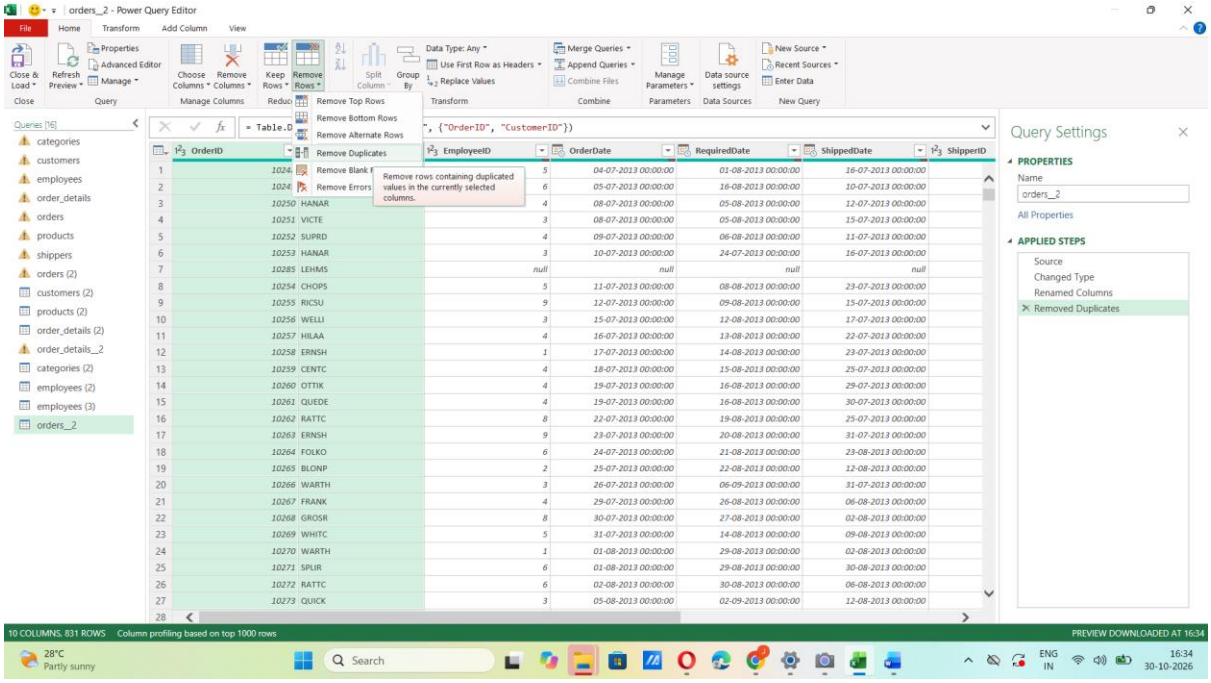
5. Employee Table

- Removed unnecessary spaces using the Trim function.
- Renamed column headers for better understanding.
- Identified Andrew Fuller as the Vice President of Sales and the top-level employee. Since he does not report to another employee, his ReportsTo value is null.

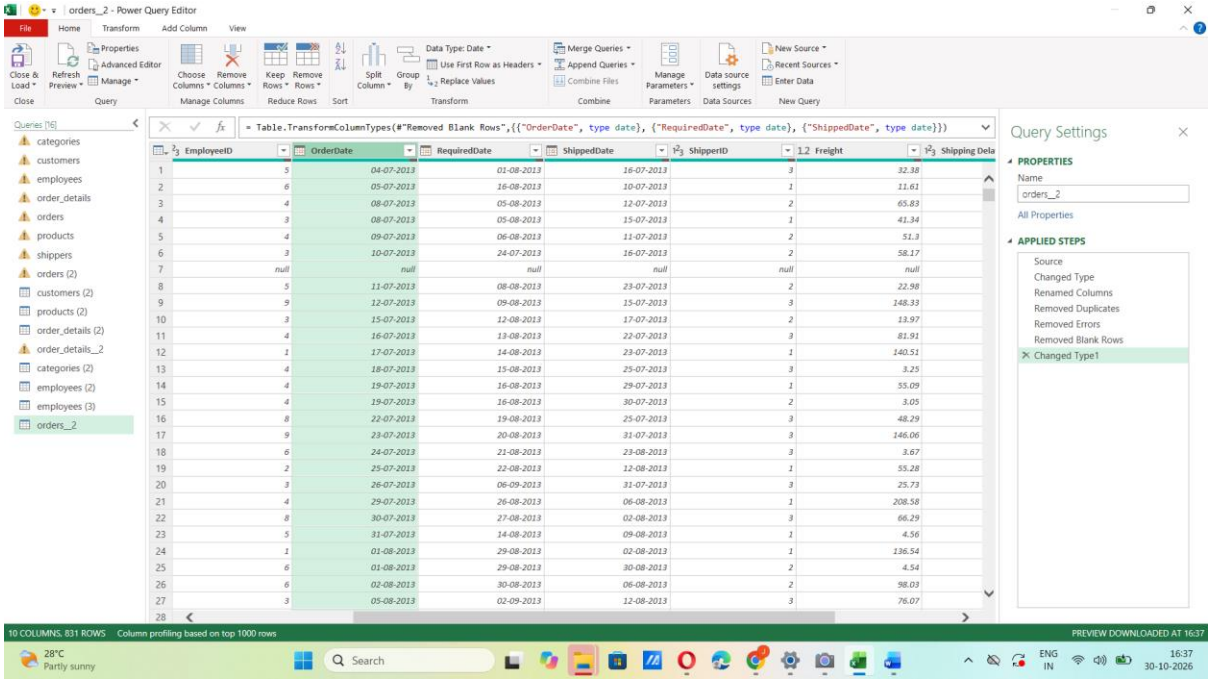


6. Cleaning Orders table - Removed duplicated with same order id and customer id,

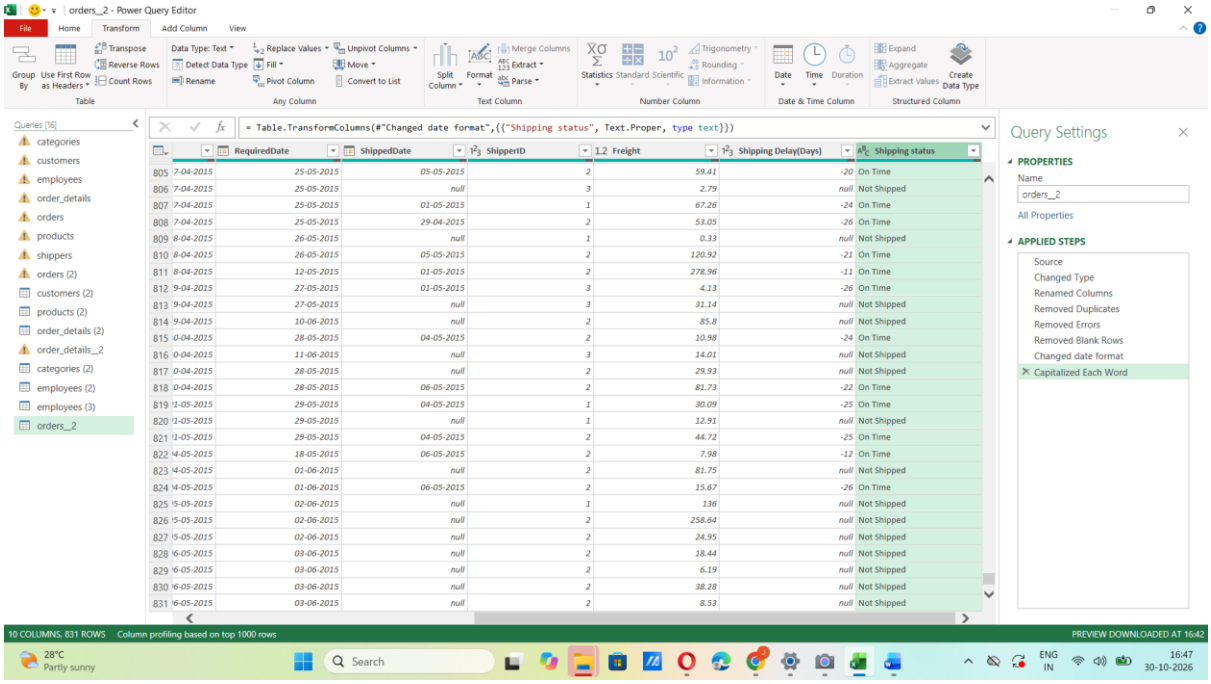
Renamed the heading, changed the data type.



Changed date format from Date/Time to Date.

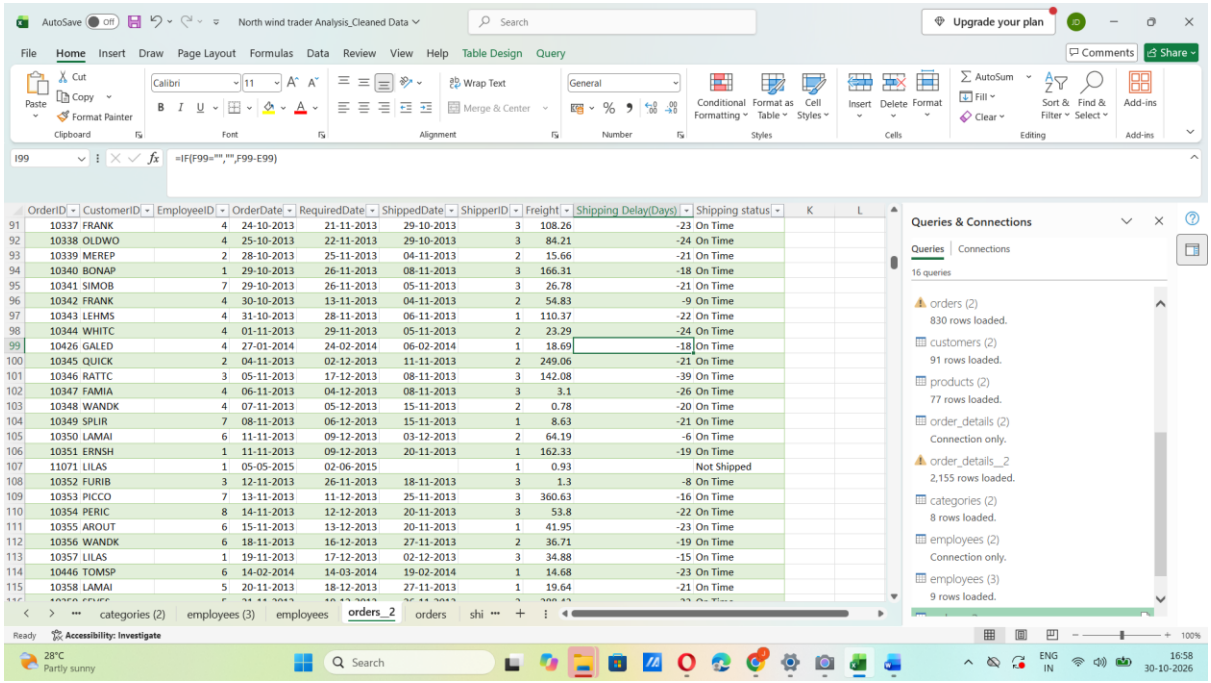


Changed text into proper format.



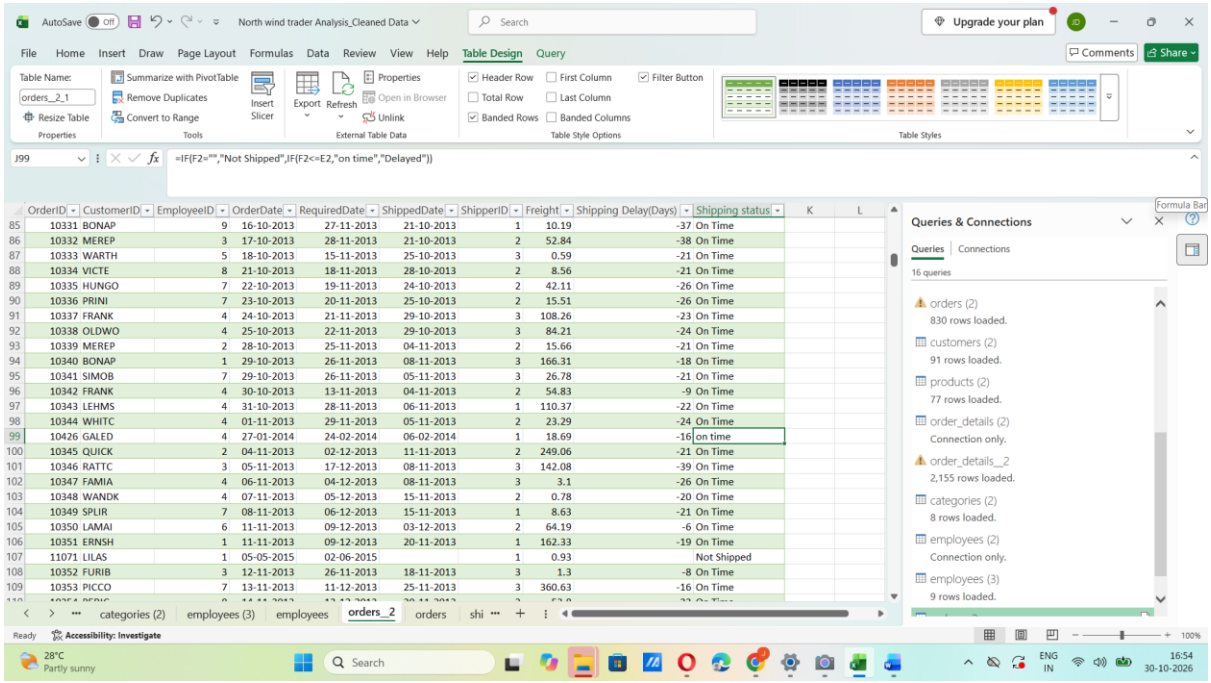
Replacing null values from shipping delay using if condition

If(f99="","",f99-e99)



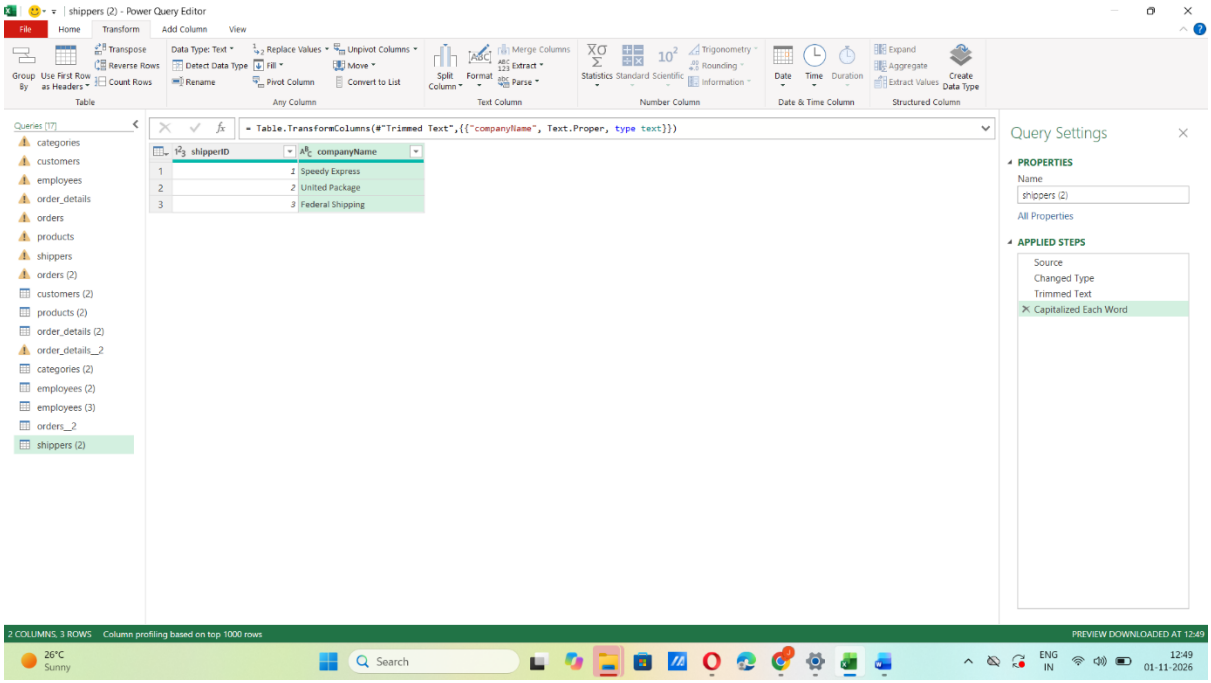
Replacing null using nested if using below formula.

=IF(F2="","Not Shipped",IF(F2<=E2,"On Time","Delayed"))



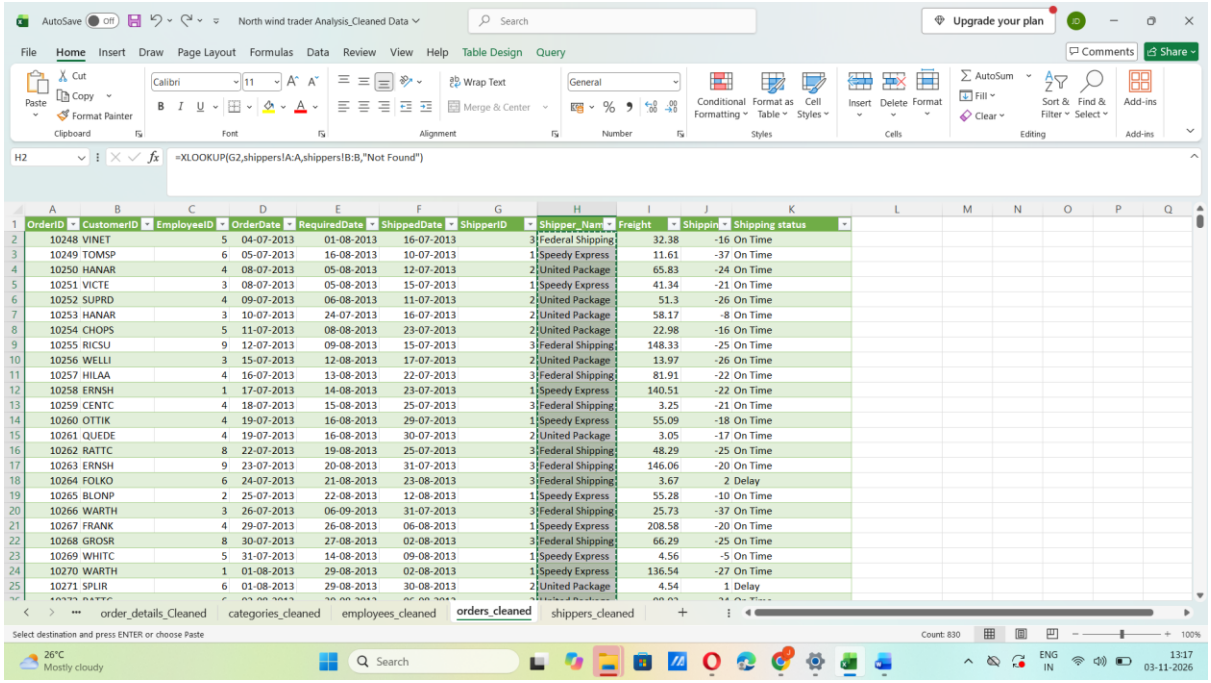
7. Shippers Table

- Removed unnecessary spaces using the Trim function.
- Applied Proper Case formatting to text values.



8. Using XLOOKUP

- Used the XLOOKUP function to retrieve shipper company names from the Shippers sheet into the Orders table using Shipper ID.
- Deleted the Shippers sheet after retrieving the shipper names to reduce the number of sheets.



Data Cleaning and Transformation Steps

1. Removed unnecessary spaces from column names and renamed columns for better understanding.
2. Applied Trim and Clean functions to remove unwanted spaces and non-printable characters.
3. Checked for duplicate records and verified unique ID columns.
4. Identified and handled missing or blank values in important columns.
5. Corrected data types for date, text, currency, and numerical columns.
6. Standardized inconsistent text formats and values.
7. Used XLOOKUP to bring shipper company names into the Orders sheet using Shipper ID.
8. Created a Sales Amount column using Unit Price, Quantity, and Discount.
9. Checked order dates, shipping dates, and shipping delays for accuracy.